

Velocity Dispersion

MATAN LAGNADO¹

¹ *University of Arizona*

1. INTRODUCTION

How does velocity dispersion change through the interaction/collision between MW and M31? In the interaction between the two masses there are many stars interacting with each other causing changes in trajectories as well as velocity. After the two galaxies collide the velocity dispersion will be important to predict the outcome. 'Heating' in a galaxy refers to the chaotic movement of stars, just like gas in a box that moves more randomly when it is being heated. A greater dispersion of vertical velocity, in and out of the plane of the galaxy, refers to more heating. We would expect the vertical velocity dispersion to change dramatically during the collision because of forces acting on the stars that change direction very rapidly. By calculating the ratio of $\frac{V_{rot}}{\sigma}$ at each time step we can get an understanding of how rotational velocities and their dispersion change during the collision.

Understanding how the velocity distribution changes throughout the collision helps us understand how galaxies evolve when they interact with their neighbors. Various forces influences the velocity dispersion which describes the changes in the shape of a galaxy. A better understanding can help us understand what influences changes in vertical velocity dispersions (thickening of the galactic plane).

When a galaxy merges with another galaxy the large disturbance in each of their systems causes stars movements to change. The velocity dispersion is a way to measure the average movement of stars through a galaxy. Just as when we look at water we may not see the intricate movement of each water molecule but we can clearly tell which direction a river flows. Continuing this analogy if another river's water is diverted and connects with another river the flow rate may change abruptly with water swirling around trying to find an optimal path. When two galaxies merge there are forces that drastically change speeds and flow of where stars are moving but with each close approach of each galaxy the stars begin to settle and find equilibrium as seen in figure 1 [N. R. Stickley & G. Canalizo \(2012\)](#). These changes can have an effect on the overall structure of the galaxy as well, for example a high vertical velocity dispersion is supports a thickened disk model [J. J. Dalcanton et al. \(2023\)](#).

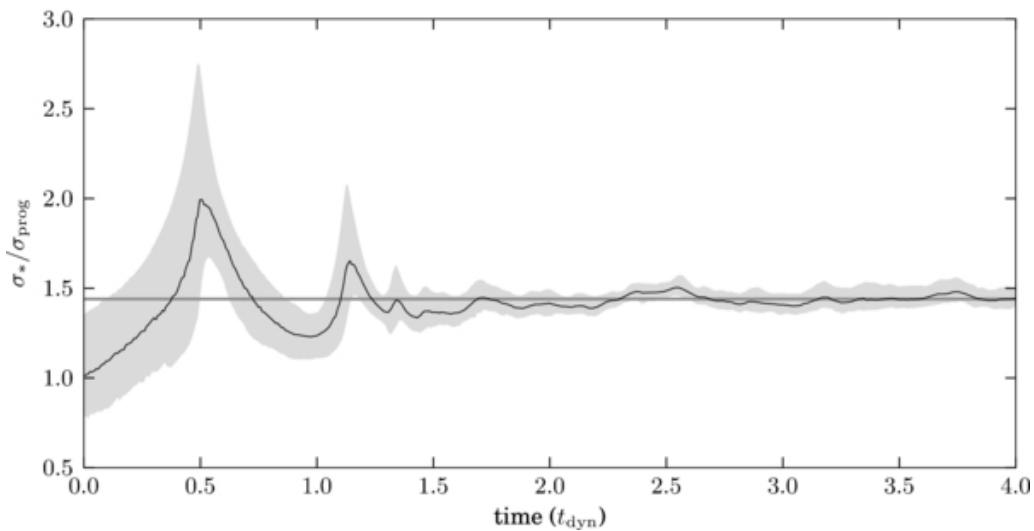


Figure 1. Shows how velocity dispersion changes throughout a merger. Black line is the mean, with uncertainty within the shaded gray region. The gray line is the final mean value of the dispersion. [N. R. Stickley & G. Canalizo \(2012\)](#)

Many questions on this topic relate to creating more accurate and higher resolution models to describe and predict velocity dispersion in galaxies [S. Han et al. \(2024\)](#). To create more accurate models more data is required which will come with the completion of the Vera C. Rubin Observatory [Pearson, W. J. et al. \(2024\)](#). Following more accurate models higher resolution models can be constructed to consider details such as dust absorption as well as 3D velocity dispersions [S. Han et al. \(2024\)](#).

2. PROPOSAL

How does the velocity dispersion of each galaxy(MW/M31) evolve throughout the interaction? How can the velocity dispersion be compared to the present day(pre-collision)?

How will you approach the specific question using the simulation data? Define all relevant equations and terms. Here you should outline the codes you'd need to write - each question will need a unique code solution. This can be described in general terms, but all steps need to be outlined (including what particle types/properties will you select and how you will select them, and specify which snapshots will you use).

The most important calculation in this question is the velocity dispersion σ . It can be calculated just like a regular standard deviation, first calculating the mean velocity and then calculating the squared average deviation from the mean in each coordinate direction:

$$V_{mean} = \frac{1}{N} \sum_{i=1}^N v_i \quad (1)$$

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (v_i - V_{mean})^2} \quad (2)$$

Because I am curious as to how it changes with respect to time I need to write code that calculates velocity dispersion at each time step, and should plot these changes. I would only look at Bulge and Disk type particles because although we have access to the Halo particles in the simulation we can't physically measure these with observation so there is no way to verify the results. The code would first calculate the mean velocity using eq. 1 and then the dispersion with eq. 2. The velocity of all Bulge and Disk particles will be used in these calculations however the Halo particles will still be used in center of mass calculations. Similarly to Homework 6 all the snapshots will be used to understand changes throughout the collision. I will be able to plot the dispersion as a function of time as shown in figure 1 [N. R. Stickley & G. Canalizo \(2012\)](#). Additionally it may be interesting to look at how dispersion changes with time in different areas of a galaxy. This can be done by binning particles based radial distance from and angle around the center of mass as shown in figure 2c [S. Han et al. \(2024\)](#).

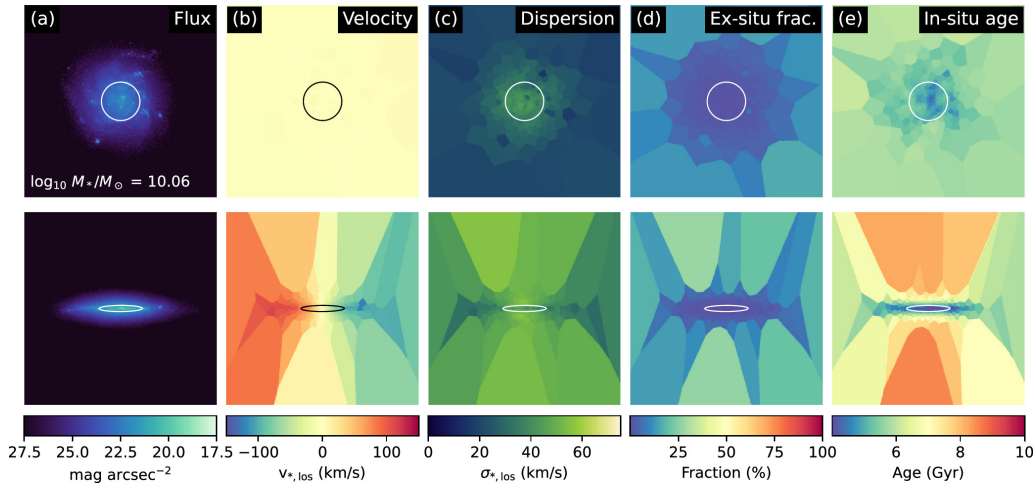


Figure 2. Figures show face-on and edge-on projections of (a) flux, (b) velocity, (c) dispersion, (d) ex situ fraction, and (e) age of in situ stars.

54 Based on the work of Stickley et al. (2012) It is suggested by his N-body simulations that the velocity dispersion
 55 increases when a galaxy merger occurs. As figure one shows the final equilibrium of the velocity dispersion increases
 56 from the value at $t = 0$. I predict that this is happening because of the addition of mass and energy into a system.
 57 Similarly to a person sitting in a chair if you give them extra weights or push the chair they are spinning in, their
 58 direction and speed are subject to change.

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